

Ders # 6

Veri Bütünlüğü Programlama ve Güvenlik

From Stanford's The complete Book

Sciore's textbook, Ch 9-10

From Raghu&Johennes DB textbook

Outline:

- General Constraints as Assertions
- Triggers
- Security

Objective:

- ☐ Specification of more general **constraints** via assertions, triggers
- ☐ Coordination of users access to data. Authorization issues on databases.

Constraints as Assertions

- General constraints: constraints that do not fit in the basic SQL categories (*such as unique, primary key, not null, foreign key..*)
 - Mechanism: **CREAT ASSERTION**
 - a constraint name,
 - followed by CHECK,
 - followed by a condition
- “The salary of an employee must not be greater than the salary of the manager of the department that the employee works for”

```
CREAT ASSERTION SALARY_CONSTRAINT
CHECK (NOT EXISTS ( SELECT *
                    FROM EMPLOYEE E, EMPLOYEE M, DEPARTMENT D
                    WHERE E.SALARY > M.SALARY AND
                          E.DNO=D.NUMBER AND
                          D.MGRSSN=M.SSN) )
```

Examples on student record database:

```
STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)
```

```
create assertion SmallSections
check (not exists
      select e.SectionId
      from ENROLL e
      group by e.SectionId
      having count(e.EId) > 30)
```

Figure 5-1

The SQL specification of the integrity constraint
“No section can have more than 30 students”

Examples on student record database:

```
STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)
```

```
create assertion ValidGradYear
check (not exists
      select s.*
      from STUDENT s
      where s.GradYear < 1863)
```

Figure 5-2

The SQL specification of the integrity constraint
“A student’s graduation year must be at least 1863”

Examples on student record database:

```
STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)
```

```
create assertion NoTakeTwice
check (not exists
      select e.StudentId, k.CourseId
      from SECTION k, ENROLL e
      where k.SectId=e.SectionId
      group by e.StudentId, k.CourseId
      having count(k.SectId)>1)
```

Figure 5-3

The SQL specification of the integrity constraint
“Students can take a course at most once”

SQL Triggers

- Objective: to monitor a database and take initiate action when a condition occurs
- Triggers are action that fire automatically based on these conditions
- Triggers are expressed in a syntax similar to assertions and include the following:
 - Event : Such as an insert, deleted, or update operation
 - Condition : optional
 - Action:To be taken when the condition is satisfied
- CREATE TRIGGER <name>: Creates a trigger
- ALTER TRIGGER <name> : Alters a trigger (assuming one exists)
- CREATE OR ALTER TRIGGER <name>
 - Creates a trigger if one does not exist
 - Alters a trigger if one does exist

implementing a trigger

Can be CREATE or ALTER

Can be BEFORE, AFTER,
INSTEAD OF

CREATE TRIGGER *TrigName*
AFTER INSERT ON *TblName*
FOR EACH ROW

Can be INSERT,
UPDATE, DELETE

WHEN (*condition*)

The condition

.....
.....

The action (like
UPDATE,
DELETE,..)

Valid Trigger Types

Triggered by INSERT:

- BEFORE INSERT statement-level.
- BEFORE INSERT row-level.
- AFTER INSERT statement-level.
- AFTER INSERT row-level.

Triggered by UPDATE:

- BEFORE UPDATE statement-level.
- BEFORE UPDATE row-level.
- AFTER UPDATE statement-level.
- AFTER UPDATE row-level.

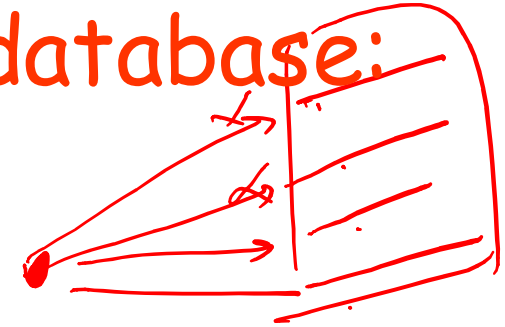
Triggered by DELETE:

- BEFORE DELETE statement-level.
- BEFORE DELETE row-level.
- AFTER DELETE statement-level.
- AFTER DELETE row-level.

To replace the triggering event:

- INSTEAD OF statement-level.
- INSTEAD OF row-level.

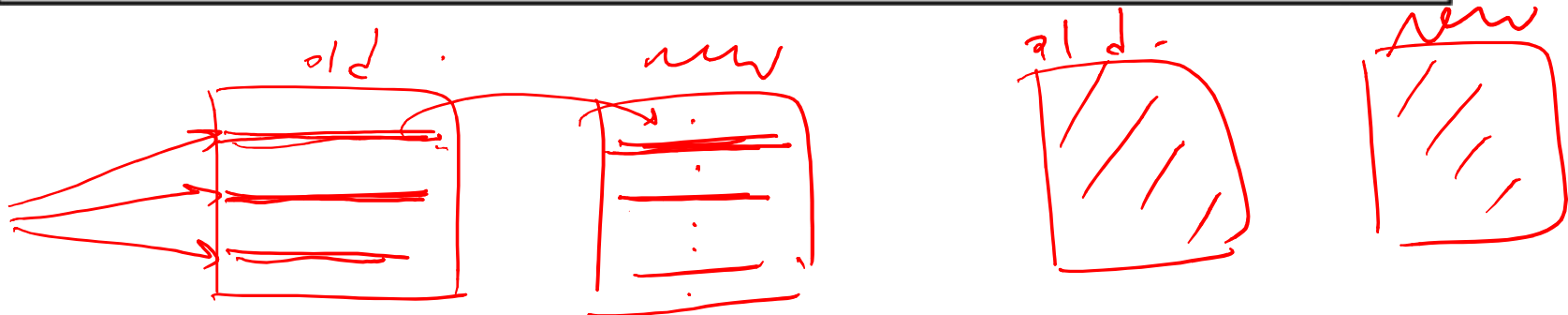
Examples on student record database:



```
create trigger LogGradeChange
after update of Grade in ENROLL
for each row
referencing old row as oldrow, new row as newrow
when oldrow.Grade <> newrow.Grade
insert into GRADE_LOG(UserName, DateChanged, EId,
                      OldGrade, NewGrade)
values(current_user, current_date, newrow.EId,
       oldrow.Grade, newrow.Grade)
```

Figure 5-4

An SQL trigger that logs changes to student grades



Examples on student record database:

```
create trigger FixBadGradYear
after insert in STUDENT
for each row
referencing new row as newrow
when newrow.GradYear > 4 + extract(YEAR, current_date)
set newrow.GradYear = null
```

Figure 5-5

An SQL trigger that replaces inappropriate graduation years with nulls

Triggers on Simplified COMPANY DB

- EMPLOYEE (NAME, SSN, SALARY, DNO, SUPERVISOR_SSN)
- DEPARTMENT (DNAME, DNO, TOTAL_SAL, MANAGER_SSN)
- **EXAMPLE 1:** A trigger to compare an employee's salary to his/her supervisor during insert or update operations:

```
CREATE TRIGGER INFORM_SUPERVISOR
BEFORE INSERT OR UPDATE OF
    SALARY, SUPERVISOR_SSN ON EMPLOYEE
FOR EACH ROW
    WHEN
        NEW.SALARY > (SELECT SALARY FROM EMPLOYEE
                       WHERE SSN=NEW.SUPERVISOR_SSN))
    INFORM_SUPERVISOR (NEW.SUPERVISOR_SSN,NEW.SSN);
```

Triggers on Simplified COMPANY DB

EMPLOYEE (NAME, SSN, SALARY, DNO, SUPERVISOR_SSN)

DEPARTMENT (DNAME, DNO, TOTAL_SAL, MANAGER_SSN)

•keep updated the DEPARTMENT.TOTAL_SAL

4 events runs the trigger:

- T1: Total_sal1 (*When a new employees is added to a department, modify the Total_sal of the Department to include the new employees salary*)
 - This trigger will execute **AFTER INSERT ON** Employee table
 - It will do the following FOR EACH ROW
 - WHEN NEW.Dno is NOT NULL
 - The trigger will UPDATE DEPARTMENT
 - By SETting the new Total_sal to be the sum of
 - » old Total_sal and NEW. Salary
 - » WHERE the Dno matches the NEW.Dno;
- T2:Total_sal2: (*changing salary of existing Emp.*)
- T3: (*Deleting an employee.*)
- T4: (*Changing department of an employee.*)

Implementing T1: «adding a new Employee»

T1: CREATE TRIGGER Total_sal1
AFTER INSERT ON Employee
FOR EACH ROW

WHEN (NEW.Dno is NOT NULL)

UPDATE DEPARTMENT
SET Total_sal = Total_sal + NEW. Salary
WHERE Dno = NEW.Dno;

EVENT

CONDITION

The action

Implementing T2: «changing salary of existing Emp»¹⁰

50
50

T2: CREATE TRIGGER Total_sal2
AFTER UPDATE OF Salary ON EMPLOYEE
REFERENCING OLD ROW AS O, NEW ROW AS N
FOR EACH ROW
WHEN (N.Dno IS NOT NULL)
UPDATE DEPARTMENT
SET Total_sal = Total_sal + N.salary - O.salary
WHERE Dno = N.Dno;

Emp. ^{Salary}
~~dept.~~
 * New.
 old

dept. ~~dept.~~

T2: CREATE TRIGGER Total_sal2
AFTER UPDATE OF Salary ON EMPLOYEE
REFERENCING OLD TABLE AS O, NEW TABLE AS N
FOR EACH STATEMENT

WHEN EXISTS (SELECT * FROM N WHERE N.Dno IS NOT NULL) OR
EXISTS (SELECT * FROM O WHERE O.Dno IS NOT NULL)

UPDATE DEPARTMENT AS D

SET D.Total_sal = D.Total_sal

+ (SELECT SUM (N.Salary) FROM N WHERE D.Dno=N.Dno)

- (SELECT SUM (O.Salary) FROM O WHERE D.Dno=O.Dno)

WHERE Dno IN ((SELECT Dno FROM N) UNION (SELECT Dno FROM O));

Emp. salary. New N
 old O

x	1
x	1
x	2
x	2

x	1
x	1
x	2
x	2

dept.

dept.
dept.
dept.
dept.

(1,2)

Cont... *T2 (for each statement)*

	Salary	DNO
	200	1
	300	2

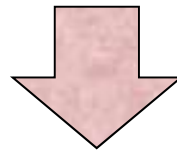
EVENT: ONLY «SAL» changes !

Old O

	Salary	DNO
	200	1
	300	2

New N

	Salary	DNO
	700	1
	900	2



ACTION !

Cont... *T2 (for each statement)*

	Salary	DNO
	200	1
	300	2

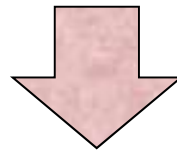
EVENT: Both «SAL» and «DNO» changes !

Old O

	Salary	DNO
	200	1
	300	2

New N

	Salary	DNO
	700	2
	900	1



ACTION !

Cont... *T2 (for each statement)*

	Salary	DNO
	200	
	300	

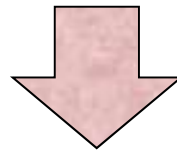
EVENT: ONLY «SAL» changes and All DNOs are NULL !

Old O

	Salary	DNO
	200	Null
	300	Null

New N

	Salary	DNO
	700	Null
	900	Null



No action !

GÜVENLİK

- **Authentication (Kimlik doğrulama)**, kullanıcının iddia ettiği kişi olduğunun ve DBMS'nin de kullanılmak istenen sistem olduğunun doğrulanması.
 - Kimlik doğrulama, belirli görevler için herhangi bir ayrıcalık/yetki vermez. **Authorization (Yetkilendirme)** için bir önkoşuldur.
- **Authorization (yetkilendirme)**; hangi kullanıcının, hangi veriler üzerinde, neler yapabileceğinin, belirlenmesi.
- **Authentication «SERVER», Authorization «SERVER», Auditing «SERVER», ...**

Authorization methods (yetkilendirme metodları)

- **Discretionary access control (*takdire dayalı*)**
 - Privileges are associated with tables
 - Based on the assumption that the users are trustworthy.
 - Privilege may propagate from one subject to another
 - Provided by SQL and flexible..
- **Mandatory access control (*zorlayıcı*)**
 - Privileges are associated with data
 - Eliminates the loopholes of discretionary method, thus suited for high-security systems (like military)
 - Less flexible

Discretionary access control

Privileges (hak, yetki)

- The table's owner grants privileges on it to other users.
- Kinds of privileges:
 - *Select, Insert, Delete, Update, References, Usage, Trigger, Execute, Under*
- SQL **GRANT/REVOKE** *statement* assigns/removes privileges to users:

SYNTAX : **GRANT** *<priv list>* **ON** *<DB element>* **TO**
<users/roles>

Example: **GRANT** *select* on **COURSE** to **public**

UNIVERSITY database

```
STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)
```

```
grant select on STUDENT to dean, admissions
grant insert on STUDENT to admissions
grant delete on STUDENT to dean
grant update on STUDENT to dean
```

```
grant select on COURSE to public
grant insert on COURSE to registrar
grant delete on COURSE to registrar
grant update on COURSE to registrar
```

```
grant select on DEPT to public
```

```
grant select on ENROLL to dean, professor
grant insert on ENROLL to registrar
grant delete on ENROLL to registrar
grant update on ENROLL to professor
```

```
grant select on SECTION to public
grant insert on SECTION to registrar
grant delete on SECTION to registrar
grant update on SECTION to registrar
```

Figure 5-6

Some SQL grant statements for the university database

users / roles

- A “user” is the person who logged into the database. The “role” is the category of users.
- For a better feasibility, privileges are granted to “roles”. Thus the admin should
 - define “roles” in the database.
 - define “users” in the database.
 - assign one or more roles to the user. *(In case of more than one role, the user is granted the union of privileges from each role)*
- The owner of an db-object (table, *view, idx,..*) automatically gets all privileges on that object.

Fine-tuning: Column privileges

GRANT *select* on ENROLL to dean, professor

- The purpose of this privilege was for keeping student enrollments and grades private.
- Instead of “all or nothing” aspect, it is better to use column privileges.
- Study the following column privileges:

GRANT *select(StudentId,grade)* on ENROLL to dean, professor

GRANT *update(Grade)* on ENROLL to professor

GRANT *insert(Sname,MajorId)* on STUDENT to admissions

Privileges required by SQL statements:

- **Select** requires “*select privilege*” on every field mentioned in the query.
- **Insert (update)** requires “*Insert (update) privilege*” for the field to be inserted (modified), **plus the “select privilege” on every field mentioned in its where clause.**
- **delete** requires “*delete privilege*” for the field to be deleted, plus the “select privilege” on every field mentioned in its where clause.

Statement	Required Privileges
select s.SId, s.SName, count(e.EId) from STUDENT s, ENROLL e where s.SId = e.StudentId and e.Grade = 'A' group by s.SId, s.SName	select(SId, SName) on STUDENT select(EId, StudentId, Grade) on ENROLL
select c.* from COURSE c where c.DeptId in (select d.DId from Dept d where d.DName = 'math')	select on COURSE select(DId, DName) on DEPT
delete from SECTION where SectId not in (select e.SectionId from ENROLL e)	select(SectId) on SECTION delete on SECTION select(SectionId) on ENROLL

Example:

Determine Who
is authorized to
execute the SQL
statements in the
table.

GRANT *update(Grade)* on ENROLL to professor

Can a professor do the followings ?
UPDATE ENROLL e SET e.Grade= 100
UPDATE ENROLL e SET e.Grade=e.Grade+1

More Fine-tuning: Privileges on **Views**

GRANT update(Grade) on ENROLL to professor

- Column privileges gives roles(users) too much power. A professor can change a grade given by another professor.
- **Solution**: Revoke the previous privileges of professors on ENROLL table. Give the privileges on the following view.

```
create view PROF_ENROLLMENTS as
  select e.*
  from ENROLL e
  where e.SectionId in
    (select k.SectId
     from Section k
     where k.Prof=current_user)

STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)

grant select          on PROF_ENROLLMENTS to professor
grant update(Grade)  on PROF_ENROLLMENTS to professor
```

Create view CurrYearGrades AS

S YearOffered, Grades

F ENROLL e, SECTION s

W e.SectionId=s.SectId and YearOffered = extract(YEAR,current_date);

GRANT select ON CurrYearGrades to DEAN

Privileges on constraints

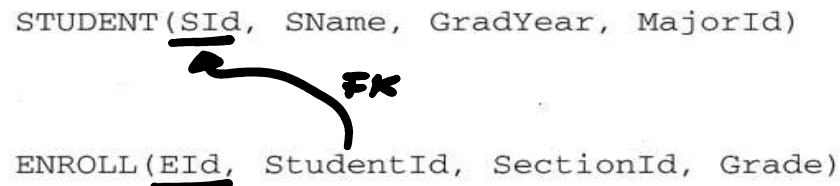
- “assertions” restricts the contents, limits what the user can do.
- It is not difficult for a malicious user to define an assertion that would make a table useless.

```
create assertion MakeTableUseless
check (not exist
      select s.*
      from STUDENT
      where SName != 'abc' and GradYear != 0)
```

- Thus, allowing to define assertions should be under authorization. We should control to use constraints (assertions, **Foreign keys**,...) that mentions STUDENT table.

GRANT references on STUDENT to dean

Reference vs. Select Privilege



User **U**: CREATE TABLE STUDENT (.....)

U: GRANT **SELECT (SId)** on STUDENT to **V**

V: CREATE TABLE ENROLL (.....PK (EId), **FK(StudentId) REFERENCES STUDENT**,
FK(SectionId)) → **REJECTED !!**

U: GRANT **REFERENCES(SId)** on STUDENT to **V**

V: CREATE TABLE ENROLL (.....PK (EId), **FK(StudentId) REFERENCES STUDENT**,
FK(SectionId)) → **ACCEPTED!!**

Why SELECT privilege is not ENOUGH ?

Remember FK enforcements: (CASCADE, SET NULL, SET DEFAULT, **NO ACTION**)

V: CREATE TABLE ENROLL (.....PK (EId), **FK(StudentId) REFERENCES STUDENT ON
DELETE NO ACTION**, FK(SectionId))

Grant-option Privileges

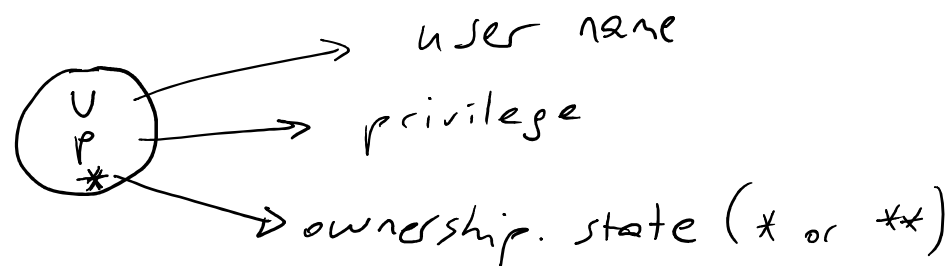
GRANT *<priv list>* **ON** *<DB element>* **TO** *<users/roles>*

GRANT insert **ON** student **TO** admission **WITH GRANT OPTION**

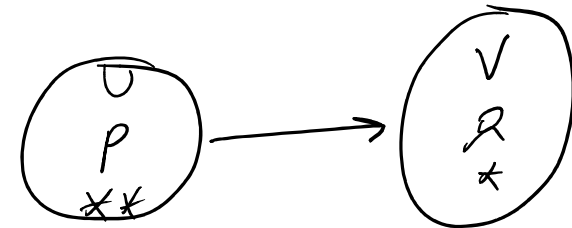
- The above grant grants the following privileges to admission:
 - To insert records on STUDENT table
 - To grant that privilege to others.
- Since it is a powerful, it is critical to grant that kind of privileges..

GRANT Diagrams

- Useful to represent «sequence of GRANTs» by a GRAPH.
- SQL system keeps track of all users privileges with «access matrices», Access Control Lists (ACL).



$\begin{cases} ** : \text{privilege is from ownership.} \\ * : \text{" is granted from ' s/w else. It includes "with grant option"} \end{cases}$



U grants Q to V
 w/ grant option
 $(Q \subset P)$

Example:

```
STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)
```

User «**admin**» grants following privileges:

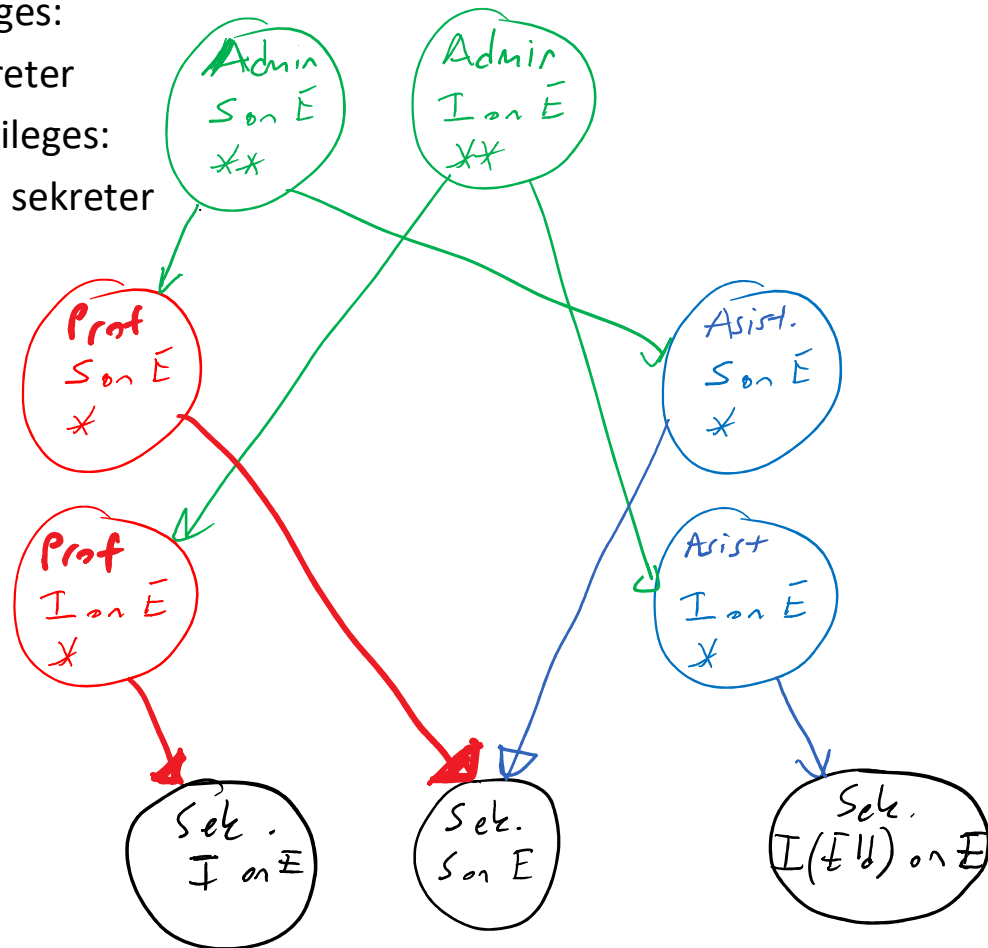
- GRANT *select,insert* ON ENROLL to **prof**,**asistan** WITH GRANT OPTION

User in «**prof**» role, grants following privileges:

- GRANT *select,insert* ON ENROLL to sekreter

User in «**asistan**» role, grants following privileges:

- GRANT *select,insert(EId)* ON ENROLL to sekreter



GRANT Diagrams/ REVOKE

REVOKE *<priv list>* **ON** *<DB element>* **FROM** *<users/roles>*
<CASCADE/RESTRICT>

CASCADE : specified privileges revoked **AND** privileges that were granted ONLY because of the granted privileges. **At last, any node that is not accessible from some ownership node is also deleted.**

RESTRICT: If the privileges that are being tried to be revoked have been passed to others, then REJECT this present REVOKE.

REVOKE GRANT OPTION FOR *<DB element>* **FROM** *<users/roles>* **<CASCADE/RESTRICT>**

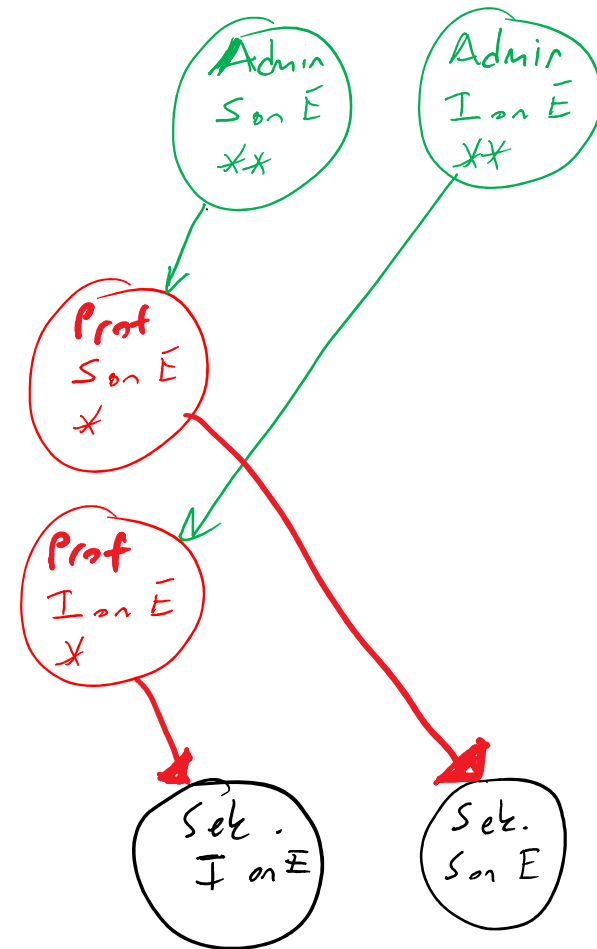
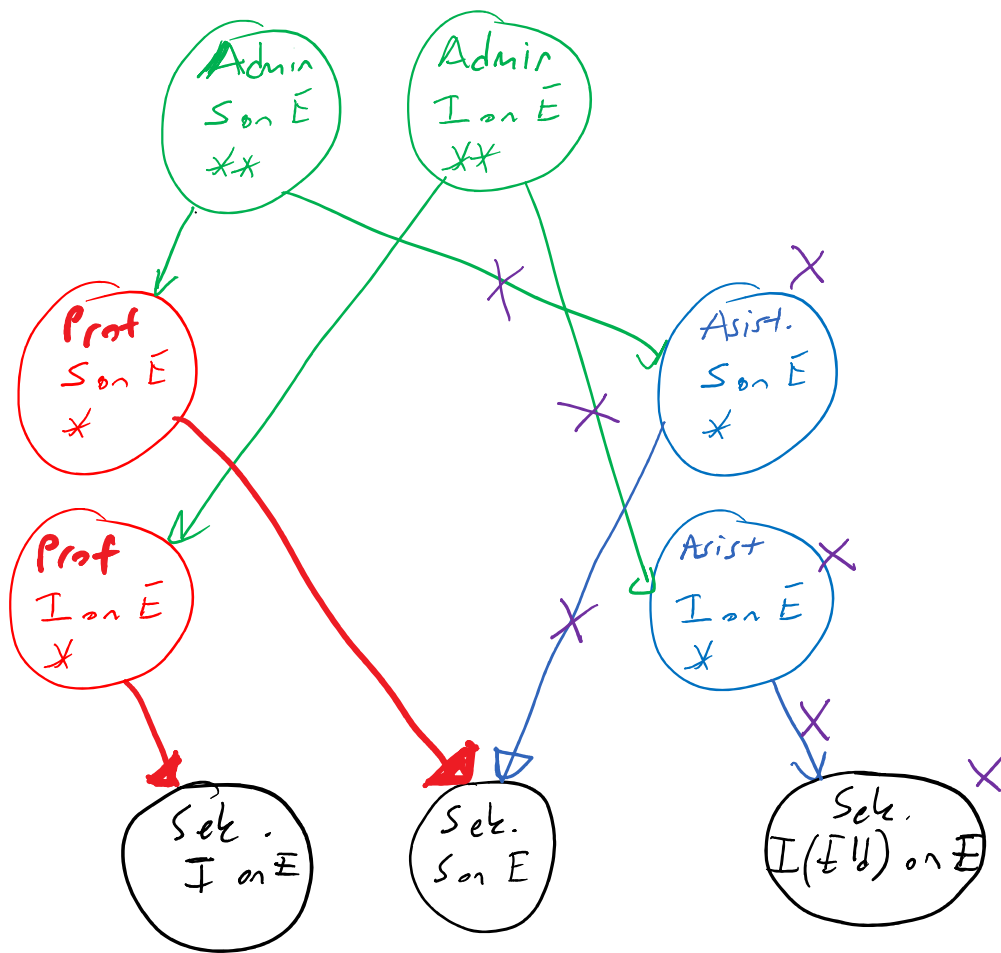
Privileges themselves remain but, the option to grant them is revoked.

Example:

STUDENT(SId, SName, GradYear, MajorId)
DEPT(DId, DName)
COURSE(CId, Title, DeptId)
SECTION(SectId, CourseId, Prof, YearOffered)
ENROLL(EId, StudentId, SectionId, Grade)

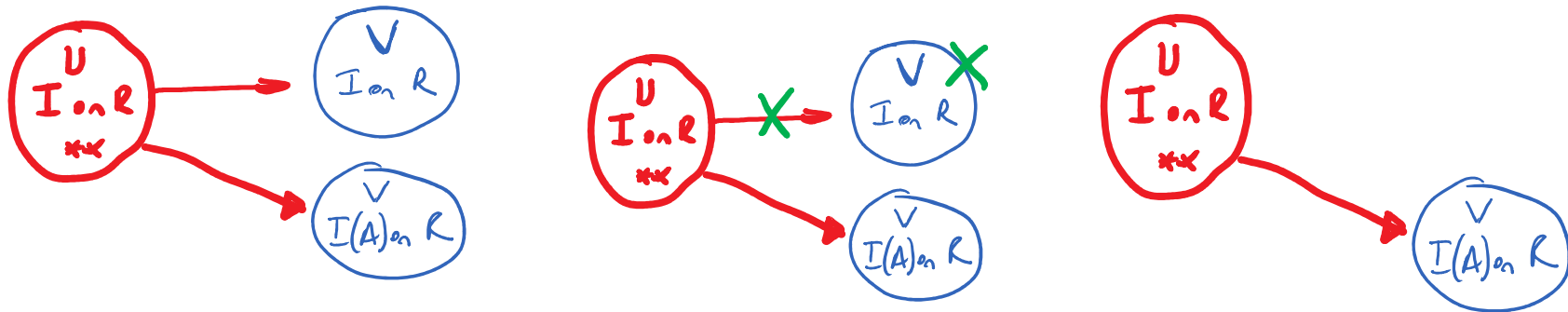
User «**admin**» grants following privileges:

- REVOKE *select,insert* ON ENROLL FROM **asistan** CASCADE



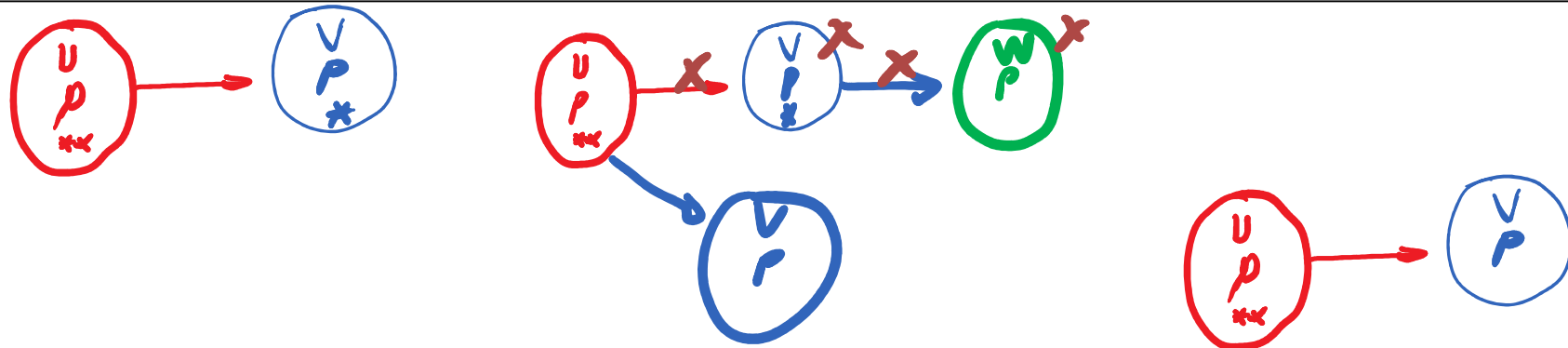
Example:

- **U**: GRANT *insert* ON R to **V**
- **U**: GRANT *insert(A)* ON R to **V**
- **U**: REVOKE *insert* ON R from **V** RESTRICT

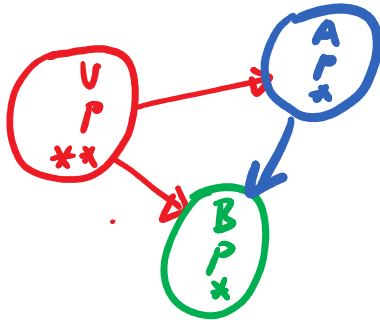


Example:

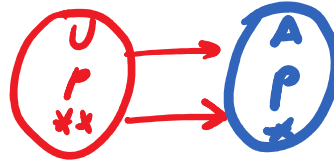
- **U**: GRANT p to **V** WITH GRANT OPTION
- **V**: GRANT p to **W**
- **U**: REVOKE **GRANT OPTION FOR** p from **V** CASCADE



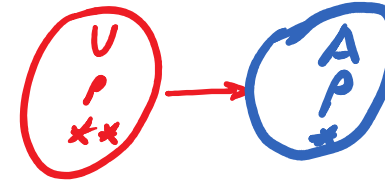
Examples:



U: REVOKE p FROM A CASCADE

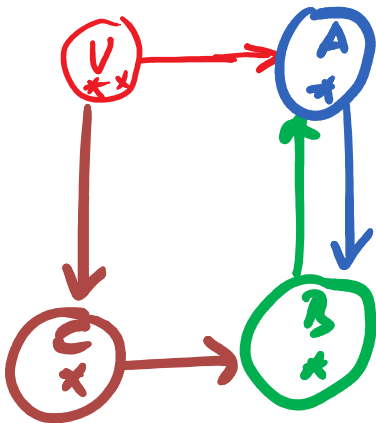


U: REVOKE p FROM A CASCADE

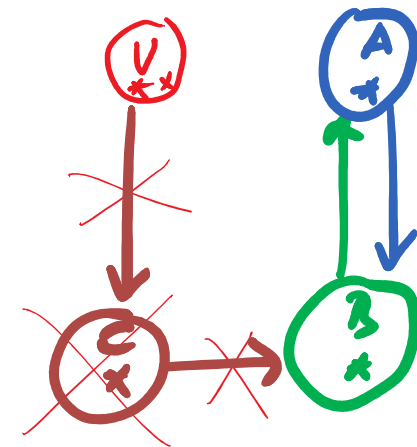
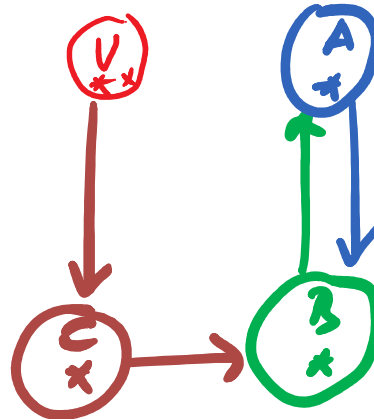


U: REVOKE **GRANT OPTION FOR** p
FROM A CASCADE

U: REVOKE p FROM A CASCADE



U: REVOKE p FROM C CASCADE



Example

- **A**: GRANT p to **B** WITH GRANT OPTION
- **A** : GRANT p to **C**
- **B** : GRANT p to **D** WITH GRANT OPTION
- **D** : GRANT p to **B**, **C**, **E** WITH GRANT OPTION
- **B** : REVOKE p from **D** CASCADE
- **A** : REVOKE p from **C** CASCADE

Show the GRANT diagram steps..

Mandatory access control

- Remember that **discretionary controls** assume that **the users are trustworthy**.
- A LOOPHOLE scenario:
- A «malicious» professor can easily violate confidentiality.

```
create table TakeAPeek(StudentId int, Title varchar(20),  
                        Grade varchar(2));
```

```
grant select on TakeAPeek to public;
```

```
insert into table TakeAPeek(StudentId, Title, Grade)  
select e.StudentId, c.Title, e.Grade  
from ENROLL e, SECTION k, COURSE c  
where e.SectionId=k.SectId and k.CourseId=c.CId;
```

- Since the professor is the owner, she can authorize anybody to look at “student grades” which is supposed to be private.
- **Mandatory access control** ensure confidentiality by assigning privileges to data, rather than to users.
- The IDEA in Mandatory access control : **When private data is copied to other tables, it still remains confidential.**
- This idea is implemented by **classification levels**. It may be built with security database software and additional discretionary SQL tools.

Security levels

- Each user, S and every object, O (table, record, ...) and is assigned a **clearances** and security level.
 - Top secret (TS)
 - Secret (S)
 - Classified (C)
 - Unclassified (UC)
- BELL-LaPADULA Model:
 - **Read Property:** S can read O; IFF $\text{class}(\text{S}) \geq \text{class}(\text{O})$: A user is authorized to READ an object «only if» his level is at least as **HIGH** as the security level of the object.
 - **Write Property:** S can write O; IFF $\text{class}(\text{S}) \leq \text{class}(\text{O})$: A user is authorized to WRITE an object «only if» his level is at least as **LOW** as the security level of the object.
(Information MUST flow from lower security level to higher)

```
create table TakeAPeek(StudentId int, Title varchar(20),
                      Grade varchar(2));
insert into table TakeAPeek(StudentId, Title, Grade)
select e.StudentId, c.Title, e.Grade
from ENROLL e, SECTION k, COURSE c
where e.SectionId=k.SectId and k.CourseId=c.CId;
```

- $\text{class}(\text{Prof}) = \text{S}$, $\text{class}(\text{TakeAPeek}) = \text{S}$
- Other users (like students, ...) $\text{class}(\text{student}) = \text{C}$. They cannot READ due to **Read prop.**

Examples:

- A LOOPHOLE scenario:

User «Tricky» : CREATE TABLE T (....)

«Tricky» : GRANT *insert* on T to Prof;

Tricky modify app code of User «Prof» that enters Student Grades by writing(copying) grades into table T.

«Prof» enters grades that is supposed to be secret. However, it is seen by «Tricky».

- In case of Mandatory Model:

class(Prof) = S → class(Application)=S

class(Tricky) = C → class(T) = C

Due to Write prop. Prof's application CANNOT write to T.

```
select c.Title
from ENROLL e, SECTION k, COURSE c
where e.SectionId=k.SectId and k.CourseId=c.CId
and e.Grade='F'
```

- Suppose ENROLL's level is *secret*, SECTION's level is *classified* and COURSE's level is *unclassified*. In such a case,

- the user that has at least a level of *secret* is able to execute the query.
- The **query's output table** has the **highest level** of its input tables = **S**

Covert Channels

- «**Inference**» of **information** about DB objects
 - that have higher security levels. (mandatory model)
 - in statistical DBs. (s.a. OLAP, Data warehouse)
 - Distributed databases. (more complex scenarios..)

Eid	SectionId	StudentId	Grade	Security Class
101	10	1234	A	S
102	11	1235	B	C

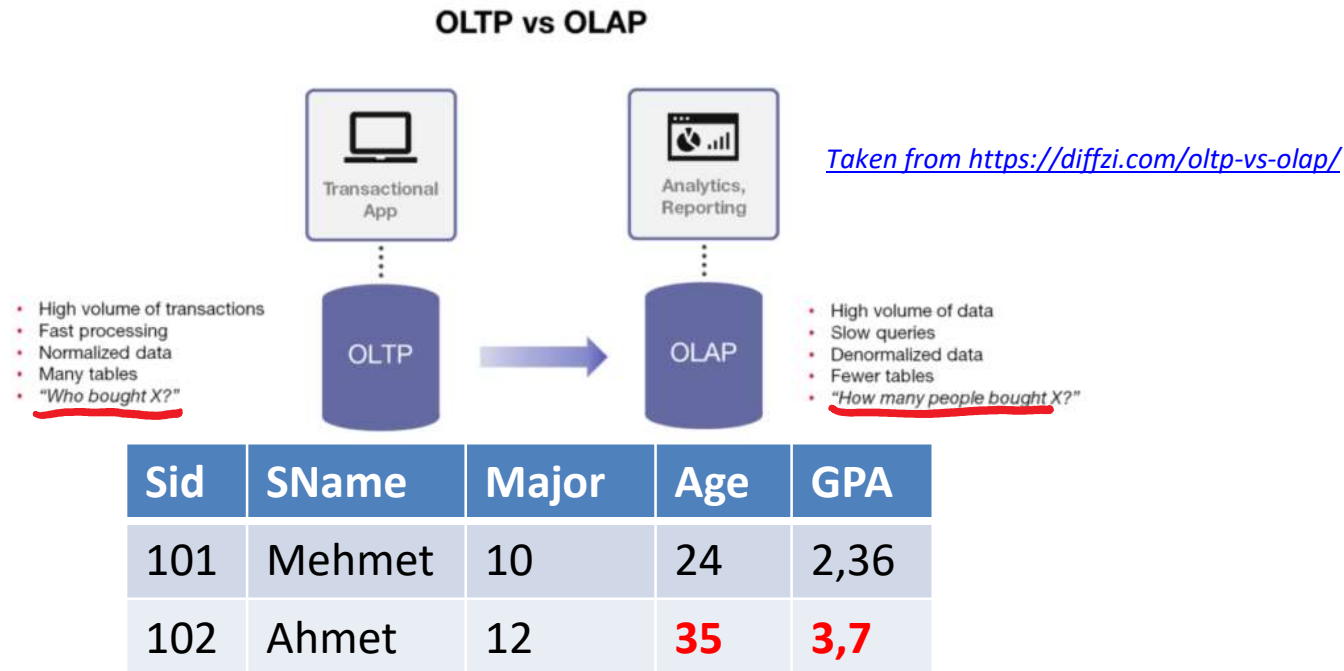
- Users with **U** clearance can see 0 tuple.
- Users with **S** or **TS** clearance can see 2 tuple.
- Users with **C** clearance can see 1 tuple.

Scenario: Users with **C** clearance: **insert <101,>**

If Accept: PK violation !

If Reject: Covert Channel !

Statistical DBs (OLAP)



- OLAP can allow **only** statistical queries.
 - Everyone knows Ahmet is the oldest student. Can «User-Tricky» know Ahmet's age, GPA ..?
 - Tricky repeatedly ask : **Q1: «How many students are there whose age is greater_equal ($\geq$) than X?» (increment X until the answer is 1) → X=35 !!**
 - Tricky ask : **Q2: «What is the max GPA of students whose age is greater_equal ($\geq$) than X=35?» → Tricky gets Ahmet's GPA=3,7**

Preventing inference in OLAP

- «Each query MUST involve at least **N** rows to run»
 - Q1 runs. But Q2 fails. Tricky is decisive, does not give up :).
 - Tricky repeatedly ask : **Q1: «How many students are there whose age is greater_equal ($\geq$) than X?» (increment X until Result is N) , Say, X=28**
 - **Q3: «What is the sum of GPA of all students whose age is greater_equal ($\geq$) than X=28?» → The result includes «Ahmet»'s GPA.**
 - **Q4: «What is the sum of GPA of all students other than «Ahmet», whose age is greater_equal ($\geq$) than X=28 and including «himself(Tricky)?»**
 - **Ahmet's GPA= Q3- (Q4-Tricky's own GPA) → Tricky gets Ahmet's GPA=3,7**

SUMMARY on Authorization methods

Preferred in commercial databases

- **Discretionary access control**
 - Privileges are associated with tables
 - Based on the assumption that the users are trustworthy.
 - Provided by SQL and flexible
- **Mandatory access control**
 - Privileges are associated with data
 - Eliminates the loopholes of discretionary method, thus suited for high-security systems (like military)
 - Less flexible